Deliverable 2: Data Exploration and Pre-processing

1. Project Title and Team Members

Project Title: Machine Learning-Driven Risk Stratification for Stroke Using Clinical and Lifestyle Factors

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2. Data Understanding

The stroke prediction dataset contains 5,110 patient records and 12 variables describing demographic, lifestyle, and clinical characteristics of individuals. The dataset consists of structured tabular clinical data where each row corresponds to a patient record and each column represents a specific feature associated with stroke risk. The target variable is stroke, indicating whether the patient experienced a stroke (1) or not (0).

The dataset includes variables such as age, gender, hypertension, heart disease, average glucose level, body mass index (BMI), smoking status, and other health-related indicators. These features represent common clinical and lifestyle factors that may influence stroke risk.

The dataset represents a binary classification problem, since the target variable contains two classes: stroke (1) and no stroke (0).

Analysis of the class distribution shows that the dataset is highly imbalanced, with 4,861 cases (95.13%) belonging to the non-stroke class and 249 cases (4.87%) belonging to the stroke class.

Data quality checks identified 201 missing values in the BMI variable, while no other variables contained missing values. Therefore, the BMI feature requires appropriate handling during the preprocessing stage, as missing values can negatively affect machine learning models.

Feature dictionary with data types:

Feature	Dtype	Feature Type	Unique Values
id	int64	Numerical	5110
gender	str	Categorical	3
age	float64	Numerical	104
hypertension	int64	Binary	2
heart_disease	int64	Binary	2
ever_married	str	Categorical	2
work_type	str	Categorical	5
Residence_type	str	Categorical	2
avg_glucose_level	float64	Numerical	3979
bmi	float64	Numerical	418
smoking_status	str	Categorical	4
stroke	int64	Binary Target	2

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to better understand the structure of the dataset, the distribution of the variables, and the potential relationships between predictors and the target variable (stroke). Numerical visualizations such as histograms, box plots, and violin plots were generated for the main numerical features (age, avg_glucose_level, and bmi) in order to examine their distributions and identify possible differences between stroke and non-stroke cases.

The analysis confirmed that stroke cases are relatively rare in the dataset, representing only 4.87% of the observations, which indicates a significant class imbalance problem. This observation highlights the importance of applying imbalance-aware preprocessing techniques during model development.

Several notable differences between stroke and non-stroke cases were observed in the numerical variables. The mean age of patients who experienced a stroke (67.73 years) is substantially higher than that of non-stroke patients (41.97 years), suggesting that age is a strong predictor of stroke occurrence. Similarly, the average glucose level is higher among stroke cases (132.54) compared to non-stroke cases (104.80), indicating a possible association between elevated glucose levels and stroke risk. The body mass index (BMI) is also slightly higher for stroke patients (30.47) compared to non-stroke patients (28.82), although the difference is less pronounced.

Correlation analysis further supports these observations. The strongest correlation with the stroke variable is observed for age (0.245), followed by heart_disease (0.135), avg_glucose_level (0.132), and hypertension (0.128). Although these correlations are moderate, they are consistent with known clinical risk factors.

These findings align with established medical knowledge that age, hypertension and cardiovascular disease significantly increase stroke risk. Age is widely recognized as one of the most significant risk factors for stroke, as the prevalence of vascular conditions increases with aging. In addition, elevated glucose levels are often associated with metabolic disorders such as diabetes, which has been shown to increase stroke risk. Hypertension and heart disease are also well-established cardiovascular risk factors reported extensively in clinical studies (O'Donnell et al., 2016).

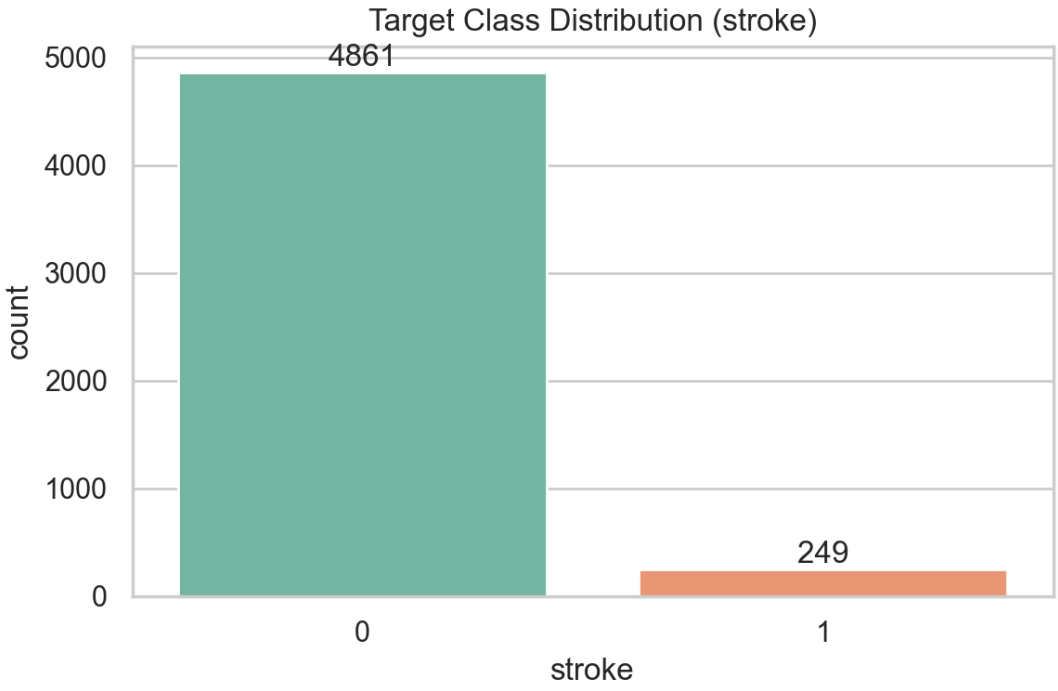

Figure 1: Target class distribution

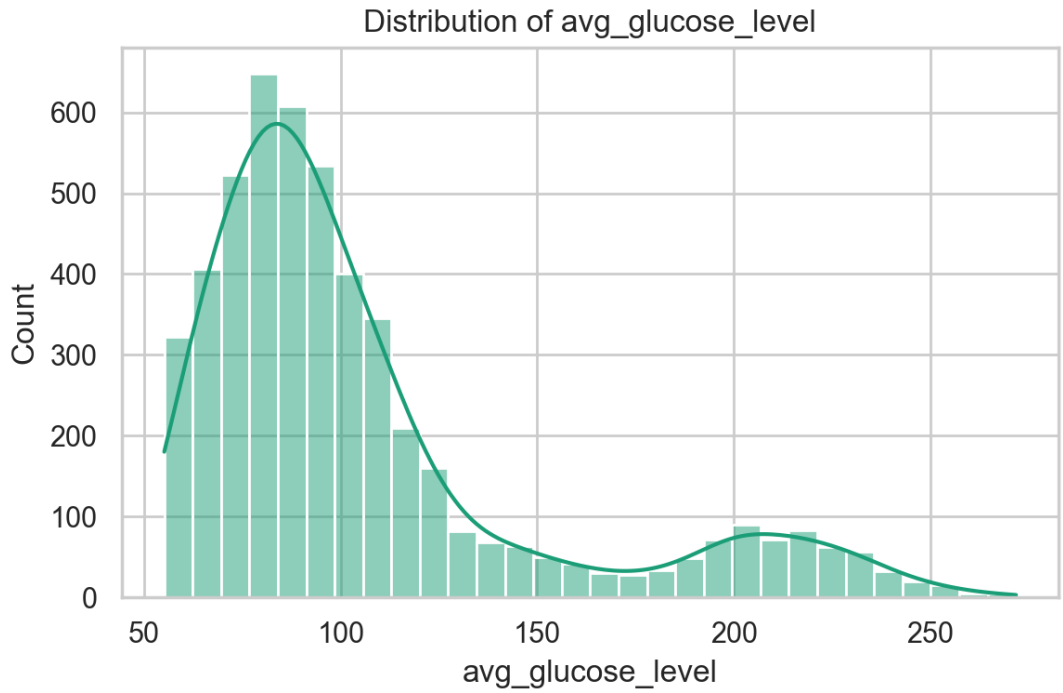

Figure 2: Histogram of avg_glucose_level

Figure 3: Avg glucose by stroke status

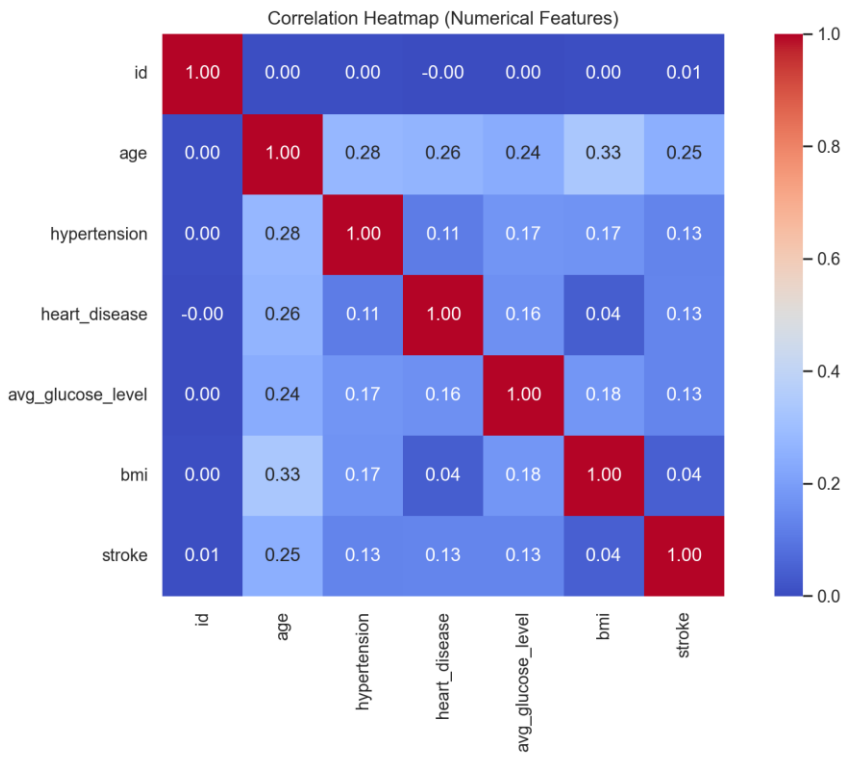

Figure 4: Correlation heatmap

4. Data Pre-processing

- Drop the identifier column (id) before modeling because it has no clinical predictive value.
- Impute missing numerical values (especially bmi) using median strategy; impute categorical values using most frequent category.
- Encode categorical features using one-hot encoding to avoid artificial ordinal relationships.
- Scale numerical features with StandardScaler for scale-sensitive algorithms (KNN, SVM, Logistic Regression), since these algorithms are sensitive to differences in feature magnitude and require normalized input features for stable model performance.
- Apply SMOTE on the training split only to address class imbalance and avoid data leakage (Chawla et al., 2002). The test set remains unchanged so that the evaluation of machine learning models reflects the true class distribution of the dataset.
- Test PCA as an experimental feature extraction step in one dataset version.

5. Pre-processed Dataset Versions

To evaluate the effect of different preprocessing strategies, multiple versions of the dataset were created. Each version applies additional data preparation steps so that their impact on model performance can later be compared during the machine learning training phase.

Version	Core Preprocessing	Scaling	Imbalance	Feature Reduction	Purpose
V1	Median/Most-frequent imputation + One-Hot Encoding	No	No	No	Baseline pipeline
V2	Same as V1	StandardScaler	No	No	Assess scaling impact
V3	Same as V2	StandardScaler	SMOTE (train only)	No	Address class imbalance
V4	Same as V3	StandardScaler	SMOTE (train only)	PCA (95% variance)	Assess dimensionality reduction

6. Selected Machine Learning Algorithms

Machine learning models have been increasingly applied in healthcare prediction and clinical decision support systems (Rajkomar et al., 2019). Several classification algorithms were selected in order to compare different modeling approaches and evaluate their suitability for stroke prediction using clinical and lifestyle data.

Logistic Regression: A widely used statistical classification model that estimates the probability of a binary outcome. It serves as a strong baseline model in medical prediction tasks due to its interpretability and probabilistic outputs, allowing clinicians to understand how individual risk factors influence stroke probability.

K-Nearest Neighbors (KNN): A distance-based classification algorithm that assigns class labels based on the majority class among the nearest neighbors in the feature space. This method helps identify local patterns among patients with similar clinical characteristics and is useful for evaluating the effect of feature scaling.

Random Forest: An ensemble learning method that combines multiple decision trees to improve predictive performance and reduce overfitting. It is particularly suitable for tabular clinical datasets because it can capture nonlinear relationships and interactions between different patient risk factors.

Support Vector Machine (SVM): A powerful classification algorithm that identifies an optimal decision boundary maximizing the margin between classes. SVM is effective for complex classification problems and often performs well when the relationship between risk factors and stroke occurrence is not strictly linear.

7. References

1. Chawla, N.V., Bowyer, K.W., Hall, L.O. and Kegelmeyer, W.P. (2002) 'SMOTE: Synthetic Minority Over-sampling Technique', *Journal of Artificial Intelligence Research*, 16, pp. 321–357.

2. Deo, R.C. (2015) 'Machine learning in medicine', *Circulation*, 132(20), pp. 1920 1930.

3. Fedesoriano (2021) Stroke Prediction Dataset. Kaggle. Available at: https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset (Accessed: 2 March 2026).

4. O'Donnell, M.J. et al. (2016) 'Global and regional effects of potentially modifiable risk factors associated with acute stroke in 32 countries (INTERSTROKE): a case-control study', *The Lancet*, 388(10046), pp. 761–775.

5. Pedregosa, F. et al. (2011) 'Scikit-learn: Machine Learning in Python', *Journal of Machine Learning Research*, 12, pp. 2825–2830.

6. Powers, W.J. et al. (2019) 'Guidelines for the Early Management of Acute Ischemic Stroke', *Stroke*, 50(12), pp. e344–e418.

7. Rajkomar, A., Dean, J. and Kohane, I. (2019) 'Machine learning in medicine', *New England Journal of Medicine*, 380(14), pp. 1347–1358.